

SMART INDIA HACKATHON 2026

TECHNICAL REPORT

Real-Time Identification of Fraud-Linked Cryptocurrency Exchanges from Victim-Reported Suspect Wallet Addressesthrough Automated Blockchain Analytics

Proposed Idea: AI-Powered Real-Time Crypto Fraud Attribution & Investigation Intelligence Platform

Problem Statement ID	SIH26183
Theme	Blockchain & Cybersecurity
Category	Software
Organization	Ministry of Home Affairs
Department	Indian Cyber Crime Coordination Centre (I4C), CIS Division
Team Name	Team Varanasi
Team ID	[INSERT TEAM ID]
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Prepared using the supplied SIH problem statement, Team Varanasi idea presentation, and the provided reference report structure.

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1. Administrative Overview

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Field	Details
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Proposed Idea Title	AI-Powered Real-Time Crypto Fraud Attribution & Investigation Intelligence Platform
Team Name	Team Varanasi

Official Problem Background

Cyber fraud victims increasingly report suspect cryptocurrency wallet addresses used by fraudsters to collect or move proceeds of crime. Such cases may arise from investment scams, task-based frauds, sextortion, ransomware, phishing, darknet-related transactions and other organized cyber-enabled financial crimes. The reported addresses are frequently non-custodial wallets, temporary burner wallets or intermediary wallets used for layering and laundering.

The investigation challenge is not simply to view a blockchain transaction. Investigators must quickly understand how funds move across multiple wallets and, where possible, identify the nearest exchange or Virtual Asset Service Provider (VASP) receiving funds from the suspect transaction path. Delays in attribution can reduce the opportunity for preservation of evidence, coordination with service providers and lawful freezing or recovery processes.

Manual tracing is technically demanding and becomes more difficult when fraud proceeds move through multiple blockchain ecosystems, bridges, DeFi protocols, mixers/tumblers or other privacy-enhancing mechanisms. The proposed solution therefore focuses on automating fund-flow tracing and converting a victim-reported wallet address into investigation-ready intelligence.

Official Problem Statement

Develop a software platform that can ingest victim-reported suspect cryptocurrency wallet addresses, automatically trace blockchain fund flows, identify associated or nearest receiving exchanges/VASPs, detect intermediary and cross-chain movement patterns, categorize wallet risk and generate actionable intelligence, alerts and standardized investigation reports for law-enforcement agencies.

Official Expected Solution Features

Real-time blockchain intelligence generation from suspect wallet addresses and transaction data.
Automated tracing of wallet-to-wallet fund movement.
Automated VASP/exchange identification where supported by reliable attribution data or clustering evidence.
Blockchain transaction graph analysis and fund-flow visualization.
Detection of intermediary wallets and suspicious laundering patterns.
Identification and continuation of supported cross-chain fund movement.
Integration capability with NCRP, SAHYOG and other authorized law-enforcement workflows.
Automated alert generation and risk categorization of wallets.
AI/ML-assisted risk detection and pattern recognition for fraud typologies.
Scalable blockchain indexing, APIs and analytics dashboards.
Generation of standardized investigation-ready reports.

2. Abstract & Problem Solution

Executive Summary

The proposed system is an AI-powered Real-Time Crypto Fraud Attribution & Investigation Intelligence Platform designed to assist law-enforcement investigators in tracing cryptocurrency fraud proceeds from a victim-reported suspect wallet address. The platform collects transaction data from supported blockchain ecosystems, constructs a transaction graph, performs hop-by-hop fund-flow analysis, detects intermediary and suspicious movement patterns, identifies known or probabilistically attributed VASP/exchange entities, flags cross-chain transitions and generates explainable risk scores.

The solution is designed around an investigator workflow rather than a generic blockchain explorer. A suspect wallet address enters through a complaint or investigator interface; the backend validates the chain and address, retrieves blockchain activity, normalizes transactions and stores relationships in a graph model. Graph algorithms and wallet-clustering signals identify relevant fund paths and candidate entities. An AI/ML-assisted risk layer evaluates behavioural and structural signals such as rapid forwarding, fan-out/fan-in behaviour, transaction velocity, hop count, anomalous movement and known-risk associations.

The resulting intelligence is presented through an interactive fund-flow dashboard with alerts, confidence indicators, explanations and a standardized investigation report. The MVP can begin with major public chains such as Bitcoin, Ethereum, Tron and Polygon, while keeping the architecture extensible for additional chains and authorized intelligence sources.

Problem–Solution Fit

Problem in Current Investigation Workflow	Proposed Platform Response	Expected Investigative Value
Manual tracing requires significant blockchain expertise and time.	Automated transaction ingestion, graph construction and hop-by-hop tracing.	Faster first-pass analysis and reduced repetitive manual work.
Reported wallets may be burner or intermediary addresses rather than exchange wallets.	Graph intelligence identifies downstream entities and intermediary paths.	Improved visibility of the fund-flow path.
Fraud proceeds can move rapidly across many wallets.	Streaming/API-based monitoring and alert rules for high-risk movement.	Earlier awareness of material fund movement.
Multi-chain movement breaks simple single-chain tracing.	Cross-chain event detection and supported bridge/entity correlation.	More continuous tracing across supported ecosystems.
Investigators need actionable outputs, not raw transaction lists.	Risk scoring, explanations, entity confidence and standardized reports.	Investigation-ready intelligence and prioritization.
Coordination with LEAs and VASPs requires structured evidence.	Exportable timelines, wallet graphs, transaction metadata and report templates.	Better evidence packaging and operational coordination.

Novelty Factor

Nearest VASP intelligence: prioritizes the closest relevant exchange/VASP encountered along a traced fraud-fund path.

Explainable risk scoring: every material score can be accompanied by visible evidence such as transaction velocity, rapid forwarding, hop count and cross-chain movement.

Graph-first investigation workflow: transactions are represented as a navigable relationship graph rather than a static spreadsheet of hashes.

Intermediary-wallet intelligence: explicitly detects addresses that appear to function as pass-through or layering nodes.

Cross-chain-aware tracing: supported bridge or cross-chain transitions are represented as investigation events instead of terminating the trace.

Pattern memory: recurring fund-flow structures can be matched over time to identify repeated fraud typologies.

LEA-ready output: complaint ingestion, alerting and standardized evidence/report generation are built around investigation workflows.

Scalable modular design: chain connectors, attribution sources and risk models can be added without redesigning the full platform.

Primary User Roles

User Role	Primary Responsibilities	Key Platform Functions
Cybercrime / LEA Investigator	Investigate victim-reported wallet addresses and trace proceeds.	Wallet search, graph exploration, risk review, alerts and evidence export.
Investigation Supervisor	Prioritize cases and review investigation quality.	Case dashboard, high-risk queue, confidence review and report approval.
Blockchain Intelligence Analyst	Validate entities and improve attribution/risk intelligence.	Entity review, clustering, label management and typology analysis.
System / Integration Administrator	Manage authorized connectors and platform operations.	API configuration, role-based access, audit controls and data retention settings.
Authorized External System	Provide or receive structured investigation data.	Complaint ingestion, alert APIs and report exchange where authorized.

3. Technical Architecture & Methodology

System Architecture

Complaint / Investigator Intake Layer – receives a suspect wallet address, transaction hash or case reference from an investigator or authorized cybercrime system.

Blockchain Data Collection Layer – validates the chain and retrieves transactions and relevant metadata through supported blockchain APIs, nodes or indexed datasets.

Normalization & Indexing Layer – converts chain-specific transaction structures into a common internal representation.

Graph Intelligence Layer – models wallets, transactions, bridges, contracts and attributed entities as connected nodes and relationships.

AI/ML Risk Layer – calculates anomaly signals, typology features, risk scores and confidence indicators.

VASP Attribution Layer – combines known entity labels, clustering evidence and direct/near-path fund-flow relationships to identify candidate exchanges/VASPs.

Investigation Intelligence Layer – merges graph, attribution and risk findings into alerts, recommendations and case evidence.

Dashboard & Integration Layer – presents interactive visualizations and exposes authorized APIs for law-enforcement workflows.

End-to-End Logical Flow:



Technology Stack

Layer	Technology / Component	Purpose in Proposed System
Frontend	React.js, Tailwind CSS, Cytoscape.js, Plotly	Responsive investigator dashboard, transaction graph visualization and analytical charts.
Backend	Python, FastAPI, REST APIs, WebSockets	Case workflows, tracing services, asynchronous processing and real-time dashboard updates.
Blockchain	Web3.py, Ethers.js, Alchemy/QuickNode, The Graph or chain-specific APIs	Transaction collection, block/contract data access and supported indexing.
Data & Indexing	PostgreSQL, custom blockchain indexer/cache	Cases, users, normalized transaction records, audit data and analytical metadata.
Graph Intelligence	Neo4j, NetworkX, Graph Data Science, wallet clustering	Fund-flow graphs, path search, centrality, communities and relationship analysis.
AI/ML	scikit-learn, XGBoost, PyTorch where justified	Anomaly detection, typology classification, risk scoring and confidence estimation.
Deployment	Nginx, Vercel, Railway or equivalent cloud infrastructure	Web serving, backend hosting and scalable deployment.
Integration	Authorized NCRP/SAHYOG/LEA APIs, report generation	Complaint ingestion, alerts and structured investigation outputs.

Step-by-Step Methodology

- Case intake: an investigator enters a suspect wallet address, supported chain information and optional case metadata.
- Address validation: the platform validates address format and chain compatibility before initiating collection.
- Blockchain data collection: transaction history and relevant metadata are retrieved through the selected provider or indexed source.
- Normalization: chain-specific transactions are transformed into a common schema containing source, destination, value, asset, timestamp, transaction hash and relevant protocol metadata.
- Graph construction: wallets and entities become nodes; transfers, bridge events and protocol interactions become relationships.
- Fund-flow expansion: the trace expands through configurable hop limits, time windows, value thresholds and risk-prioritized paths.
- Intermediary analysis: pass-through behaviour, rapid forwarding, fan-out/fan-in structures and other patterns are calculated.
- VASP/entity attribution: downstream addresses are compared against verified labels and clustering/relationship evidence to identify candidate receiving exchanges or VASPs.
- Cross-chain detection: supported bridge/protocol events are identified and correlated to continue the investigation across ecosystems where evidence is sufficient.
- AI/ML scoring: graph and transaction features are evaluated for anomaly and typology risk; outputs include a score and explanation.
- Alert and recommendation generation: high-risk movements, likely exchange destinations and uncertain paths are prioritized for investigator review.
- Investigation report generation: the system produces a standardized summary containing case inputs, trace evidence, findings, confidence and recommended next steps.

Blockchain Tracing & Graph Intelligence

The platform represents a cryptocurrency investigation as a directed, attributed graph. Wallet addresses are nodes; transfers are edges; exchanges/VASPs, bridges, contracts and other labelled services can be represented as entity nodes or attributed wallet clusters. Each edge carries evidence such as transaction hash, timestamp, asset, value and chain identifier.

A practical trace should not blindly traverse every reachable transaction because blockchain graphs can expand exponentially. The proposed engine therefore uses configurable controls: maximum hop depth, time windows, minimum value thresholds, asset continuity checks and risk-prioritized path expansion. These controls help investigators focus on paths that are plausibly relevant to the reported fraud.

Useful graph operations include shortest or constrained path search to labelled entities, neighborhood expansion, degree and centrality analysis, community detection, fan-in/fan-out measurement and temporal sequence analysis. The graph engine should preserve transaction evidence so that a visual relationship always links back to underlying on-chain records.

VASP Identification & Wallet Attribution

VASP identification is treated as an evidence-based attribution task rather than a guaranteed identity lookup. The strongest signal is a verified entity label from a trusted and authorized attribution source. Where direct labels are unavailable, the platform may calculate clustering and relationship evidence and present a candidate entity with a confidence level, while clearly separating verified attribution from probabilistic inference.

The 'nearest VASP' objective can be implemented as the earliest downstream attributed VASP/entity encountered on a relevant traced path, subject to path validity and confidence rules. The report should include hop count, transaction evidence, path continuity, attribution source category and confidence, allowing investigators to understand why the entity was surfaced.

Cross-Chain Analytics

Cross-chain tracing is implemented as a modular capability. The system detects supported bridge or protocol interactions, records the source-chain event and attempts to correlate the corresponding destination-chain movement using available transaction metadata, asset and amount relationships, timing and supported bridge intelligence. The system should not claim deterministic continuity where evidence is insufficient; ambiguous transitions are flagged for review with a lower confidence score.

Chain adapters isolate chain-specific collection logic.

Bridge events are normalized into cross-chain investigation events.

Trace continuity records source chain, destination chain, event timestamps and evidence.

Cross-chain paths are visualized as explicit transitions instead of hidden wallet hops.

Unsupported or privacy-obscured movement is surfaced as a trace limitation rather than fabricated certainty.

AI/ML Risk Detection & Explainability

The AI/ML layer is designed to assist prioritization, not to autonomously make legal conclusions. A feature set can combine transaction behaviour, graph structure and known-risk associations. Candidate features include transaction frequency, value movement speed, rapid forwarding ratio, hop count, fan-in/fan-out, dormant-then-active behaviour, repeated counterparties, cross-chain transitions and exposure to previously labelled high-risk entities.

For an MVP, a hybrid approach is recommended: deterministic rules for clear, explainable patterns plus anomaly detection or supervised models for more complex signals. Gradient-boosted models can provide strong tabular baselines; graph-derived features can be added from Neo4j/NetworkX. Where sufficient labelled data exists, typology classifiers can be trained. Every model output should be accompanied by feature-level reasoning, confidence and a clear statement that it is an investigative risk indicator.

Illustrative Explainable Risk Score

Signal	Illustrative Evidence	Interpretation
Rapid forwarding	Most received value forwarded shortly after receipt.	Possible pass-through/layering behaviour.
High fan-out	Funds split to many downstream wallets.	Possible fragmentation of proceeds.
High fan-in	Multiple sources converge into one wallet.	Potential aggregation point requiring review.
Cross-chain transition	Movement enters a supported bridge or chain transition.	Trace complexity increases; continuation required.
Known-risk exposure	Path reaches a labelled high-risk entity or service.	Priority increases based on attribution confidence.
Anomalous graph structure	Behaviour differs significantly from baseline patterns.	Requires investigator review, not automatic guilt conclusion.

Dashboard, Alerts & Investigation Reports

The investigation dashboard should provide a single case workspace containing the suspect wallet, chain, transaction timeline, fund-flow graph, path to candidate VASP/entity, intermediary-wallet indicators, cross-chain events, risk score and explanation. Interactive graph controls should allow investigators to filter by hop, date, asset, amount and confidence.

- High-risk fund movement alerts with configurable thresholds.
- Candidate VASP/exchange cards showing confidence and supporting evidence.
- Intermediary wallet flags and behavioural indicators.
- Cross-chain transition markers.
- Case timeline and transaction evidence table.
- AI/ML risk explanation and confidence.
- Automated investigation recommendations such as review path, preserve transaction evidence or prioritize entity contact through authorized process.
- Standardized report export containing trace methodology, evidence, limitations and findings.

4. Feasibility, Risk Analysis & Mitigation

Technical Feasibility

The primary transaction data for public blockchains is publicly observable, subject to each chain's architecture and provider access conditions.

Mature APIs, nodes and indexing services can supply transaction data for a focused MVP.

Graph databases and graph libraries are well suited to wallet-to-wallet relationship modelling.

The solution can begin with a limited number of high-priority chains such as Bitcoin, Ethereum, Tron and Polygon, then expand through modular adapters.

AI/ML can begin with explainable anomaly detection and risk scoring before attempting heavy end-to-end deep learning.

A functional prototype does not require a proprietary blockchain; the innovation is the automated investigation workflow and intelligence layer.

Financial & Operational Viability

The MVP can be developed primarily with open-source frameworks and carefully selected developer-tier blockchain APIs. The most resource-intensive elements are large-scale blockchain indexing, long-term data retention and premium attribution intelligence. These can be controlled in an MVP through bounded trace depth, caching, selective chain support and modular external connectors.

Operationally, the platform fits a case-driven workflow: a complaint or investigator provides a wallet; the system performs automated enrichment; the investigator reviews evidence and confidence; and structured findings can be exported or integrated into authorized systems. The architecture is therefore viable as a decision-support layer that complements existing cybercrime investigation processes rather than replacing them.

24-Hour MVP Scope

Priority	Scope	Implementation Approach
Must Have	Wallet input, chain validation, transaction collection, graph visualization, multi-hop tracing, basic risk rules, candidate VASP/entity display, investigation report.	Use one or two reliable chain APIs, seeded labels/demo cases and bounded hop depth.
Should Have	Intermediary wallet detection, WebSocket/near-real-time alerts, Neo4j persistence, explainable anomaly score.	Graph features + rule/ML hybrid scoring.
If Time Remains	Cross-chain bridge correlation, advanced clustering, live authorized integration, typology classifier.	Modular prototype with clearly labelled experimental status.

Risk Assessment Matrix

Risk	Likelihood	Impact	Mitigation
API quota, downtime or provider changes	Medium	Medium	Abstract provider layer; cache data; maintain fallback/demo datasets.
Incorrect or stale VASP attribution	Medium	High	Separate verified labels from inferred candidates; display source/confidence; enable analyst review.
Trace explosion in large transaction graphs	High	High	Hop/time/value limits; risk-prioritized expansion; asynchronous processing.
Cross-chain correlation ambiguity	High	High	Use explicit evidence rules; mark uncertain links; never fabricate deterministic continuity.
Sparse labelled fraud data for ML	High	Medium	Start with anomaly detection and expert rules; evaluate only when reliable labels exist.
Privacy/legal handling of case data	Medium	High	Role-based access, audit logs, data minimization, retention controls and authorized integration only.
Mixers/privacy-enhancing mechanisms	Medium	High	Surface trace limitations and risk signals; escalate for specialist analysis.
Scope creep during hackathon	High	High	Prioritize one complete end-to-end investigation journey before advanced modules.

Evaluation Plan

The following metrics are proposed for prototype evaluation. They are evaluation measures, not claimed achieved results.

Area	Proposed Metric	How It Will Be Evaluated
Tracing	Trace generation latency	Time from valid wallet submission to initial graph and findings.
Graph analysis	Relevant path discovery	Ability to surface known/seeded downstream paths in controlled test cases.
VASP attribution	Precision/coverage on labelled test set	Compare surfaced entities against verified or seeded attribution labels.
Risk model	Precision/recall, ROC-AUC or anomaly review rate as applicable	Use held-out labelled or carefully constructed evaluation data.
Cross-chain module	Correct correlation rate on supported known events	Validate against known bridge transactions where ground truth exists.
Explainability	Investigator review usefulness	Structured user testing: can the reviewer understand why a score/entity was shown?
System	API success rate and dashboard responsiveness	Measure end-to-end prototype stability under controlled load.

5. Socio-Economic Impact & Scalability

Primary Beneficiaries

Cybercrime investigators and law-enforcement agencies handling cryptocurrency-enabled fraud.
 Indian Cyber Crime Coordination Centre (I4C) and connected investigation workflows where authorized.
 Victims of cryptocurrency-enabled cyber fraud, who benefit indirectly from faster and more structured tracing.
 Financial and virtual-asset compliance teams responding to lawful preservation or information requests.
 Blockchain intelligence analysts who currently spend substantial time performing repetitive manual transaction tracing.

Quantifiable Benefits / Proposed KPIs

The supplied problem statement does not provide baseline operational measurements from which defensible percentage improvements can be claimed. Therefore, this report uses measurable KPIs instead of inventing performance numbers.

Metric	How It Will Be Measured	Desired Direction
Initial trace time	Time from wallet submission to first structured fund-flow graph.	Lower
Manual analysis steps	Number of repetitive lookup/tracing actions required per case.	Lower
Entity identification coverage	Share of relevant test cases where a labelled/candidate VASP is surfaced.	Higher
Attribution confidence calibration	Relationship between confidence score and observed correctness on labelled data.	Better calibrated
Risk review precision	Share of high-priority alerts confirmed as worthy of investigation.	Higher
Evidence completeness	Presence of transaction hashes, timestamps, path and entity evidence in exported reports.	Higher
Cross-chain continuity	Supported transitions successfully represented with sufficient evidence.	Higher
Case prioritization latency	Time required to rank a new case for investigator review.	Lower

Social and Operational Benefits

Reduces repetitive manual blockchain tracing and enables investigators to focus on interpretation and action.
 Converts victim-reported wallet addresses into structured, graph-based intelligence rather than isolated identifiers.
 Supports earlier identification of candidate exchanges/VASPs, potentially improving the timeliness of authorized preservation or freezing processes.
 Strengthens evidence organization through transaction timelines, wallet relationships and standardized reports.
 Improves coordination by exposing consistent case outputs to authorized stakeholders.
 Creates a reusable intelligence layer that can learn from recurring fraud typologies over time.

Scalability & Future Scope (2–3 Years)

Phase 1 – Focused MVP: support a small number of major chains, investigator wallet intake, transaction graph tracing, seeded/verified entity labels, explainable risk rules and report generation.

Phase 2 – Intelligence Expansion: add more chains, custom indexing, advanced graph analytics, alerting and analyst feedback loops.

Phase 3 – Cross-Chain and Typology Intelligence: add supported bridge correlation, richer entity intelligence, supervised typology models and model monitoring.

Phase 4 – Institutional Integration: connect through authorized APIs with complaint, case-management and intelligence systems; add role-based workflows and audit controls.

Phase 5 – National-scale Architecture: partitioned indexing, asynchronous trace jobs, caching, message queues, model versioning and multi-region deployment where operationally required.

Competitive Positioning

Generic blockchain explorers expose transactions but require investigators to manually interpret relationships. Commercial blockchain intelligence platforms may provide powerful attribution and analytics but can be expensive or outside the scope of a student-built prototype. The proposed SIH solution differentiates itself through its problem-specific workflow: victim-reported wallet intake, automated hop-by-hop tracing, nearest relevant VASP prioritization, intermediary-wallet detection, cross-chain event awareness, explainable risk scoring and investigation-ready reporting designed around cybercrime response.

The project should not claim that no similar blockchain analytics tools exist. Its novelty is the integration and automation of these capabilities into a focused investigation workflow aligned with the stated SIH problem requirements.

6. References & Literature Survey

A. Problem Statement / Official Sources

[1] Smart India Hackathon 2026, SIH26183 — Real-Time Identification of Fraud-Linked Cryptocurrency Exchanges from Victim-Reported Suspect Wallet Addresses through Automated Blockchain Analytics, supplied SIH problem statement.

[2] Team Varanasi, SIH2026 Idea Presentation — Technical Approach, Feasibility and Viability, Impact and Benefits, and Research and References, supplied project presentation.

B. Research / Technical References

[3] NIST — Blockchain Technology. <https://www.nist.gov/blockchain>

[4] ICFACT 2025 — Blockchain Forensics, Atlantis Press. <https://www.atlantispress.com/proceedings/icfact-25/126023943>

[5] arXiv — Blockchain Technology, Architecture, Cryptography and Cross-Chain Interoperability reference supplied in the project presentation. <https://arxiv.org/html/2505.08772v1>

[6] I4C — National Cyber Crime Reporting Portal. <https://i4c.mha.gov.in/ncrp.aspx>

C. Technical Documentation

[7] FastAPI Documentation — Python API and asynchronous web-service development.

[8] React Documentation — frontend application development.

[9] Neo4j Documentation — graph database and graph data science workflows.

[10] NetworkX Documentation — graph analysis in Python.

[11] scikit-learn Documentation — machine-learning algorithms and evaluation.

[12] Web3.py Documentation — Ethereum-compatible blockchain interaction.

[13] Ethers.js Documentation — JavaScript/TypeScript Ethereum interaction.

D. Literature Survey Summary

Reference	Area	Relevant Learning	Use in Proposed System
[3]	Blockchain fundamentals	Public ledgers provide traceable transaction histories and distributed data structures.	Foundation for transaction collection and evidence preservation.
[4]	Blockchain forensics	Transaction graphs, address clustering and suspicious-pattern analysis are central to investigations.	Supports graph-based tracing and wallet relationship analysis.
[5]	Blockchain architecture & interoperability	Cross-chain architecture introduces additional movement and attribution complexity.	Supports modular multi-chain and cross-chain design.
[6]	Cybercrime reporting ecosystem	Cybercrime complaints require structured coordination and investigation workflows.	Motivates NCRP/LEA-oriented intake and report integration.
[7]-[13]	Implementation frameworks	Modern web, graph and ML tooling can support modular investigative software.	Forms the proposed MVP technology stack.

E. Patent / Infringement Position

No patent search or formal freedom-to-operate analysis has been performed as part of this SIH technical report. Accordingly, no claim of non-infringement is made. The proposed system is described at the functional and engineering-architecture level. Any production deployment or commercialization should include formal legal, intellectual-property, data-governance and regulatory review.

Source and Evidence Discipline

Official requirements in this report are based on the SIH26183 problem statement supplied by the team. Technical choices such as the selected frameworks, graph algorithms, risk features, MVP boundaries and evaluation metrics are proposed engineering decisions informed by the supplied Team Varanasi presentation and standard software design practice. No unsupported accuracy, recovery-rate or socio-economic percentage is claimed. Entity attribution and cross-chain continuity are explicitly treated as evidence- and confidence-dependent findings.

